

Fluid Dynamics Analysis of a Square Contraction Nozzle

This document presents a comprehensive aerodynamic evaluation of a converging nozzle designed for a wind tunnel inlet. The system features a symmetric square geometry transitioning from a large inlet profile down to a restricted outlet test section, intentionally driving localized fluid acceleration.

1. Continuity Equation and Mass Conservation

To evaluate velocity scaling through the nozzle architecture, we apply the **Continuity Equation for incompressible flow**. This fundamental engineering law dictates that the total volumetric flow rate (Q) passing through an enclosed duct system must remain completely uniform at all cross-sections:

$$Q = A_1 \cdot V_1 = A_2 \cdot V_2 \quad (1)$$

Where:

- A_1, A_2 = Cross-sectional areas of the inlet and outlet (m^2)
- V_1, V_2 = Air velocities at the inlet and outlet boundary planes (m/s)

Geometric Boundary Conditions

The design specifications define a square inlet side length of 210 mm (0.21 m) and a square outlet side length of 78 mm (0.078 m). Computing the true geometric surface areas yields:

$$A_1 = 0.21 \text{ m} \times 0.21 \text{ m} = 0.04410 \text{ m}^2 \quad (2)$$

$$A_2 = 0.078 \text{ m} \times 0.078 \text{ m} = 0.006084 \text{ m}^2 \quad (3)$$

The System Contraction Ratio

The theoretical velocity amplification coefficient is established by evaluating the area dimension ratio:

$$\text{Contraction Ratio} = \frac{A_1}{A_2} = \frac{0.04410}{0.006084} \approx 7.2485 \quad (4)$$

Thus, the localized fluid velocities map directly to the geometric constraint:

$$V_2 = 7.2485 \cdot V_1 \quad (5)$$

Analytical Velocity Profile Solutions

The integrated propulsion fan generates a volumetric flow displacement rate of 54.33 CFM. Converting this imperial performance metric into SI standard unit equivalents (1 CFM = 0.00047195 m³/s):

$$Q = 54.33 \times 0.00047195 \approx 0.025641 \text{ m}^3/\text{s} \quad (6)$$

By rearranging the baseline continuity expressions ($V = \frac{Q}{A}$), the calculated velocity profiles yield:

$$V_1 = \frac{0.025641 \text{ m}^3/\text{s}}{0.04410 \text{ m}^2} \approx 0.5814 \text{ m/s} \quad (\approx 1.30 \text{ mph}) \quad (7)$$

$$V_2 = \frac{0.025641 \text{ m}^3/\text{s}}{0.006084 \text{ m}^2} \approx 4.2145 \text{ m/s} \quad (\approx 9.43 \text{ mph}) \quad (8)$$

Evaluating the differential net velocity amplification (ΔV) exclusively generated by this nozzle profile:

$$\Delta V = V_2 - V_1 = 4.2145 \text{ m/s} - 0.5814 \text{ m/s} = 3.6331 \text{ m/s} \quad (\approx 8.13 \text{ mph}) \quad (9)$$

2. Bernoulli Principle and Energy Transformations

To characterize the underlying mechanical energy transitions forcing this velocity change, we invoke the **incompressible Bernoulli Equation**. Assuming a steady, frictionless streamline flow with negligible gravitational height variations, fluid pressure energy and kinetic energy maintain global equilibrium:

$$P_1 + \frac{1}{2}\rho V_1^2 = P_2 + \frac{1}{2}\rho V_2^2 \quad (10)$$

Where P_1, P_2 isolate localized static pressure states (Pa) and ρ represents standard sea-level air density (1.225 kg/m³ at standard temperature and pressure).

Nozzle Pressure Drop Characterization (ΔP)

Isolating the localized pressure differential parameter across the converging nozzle channel yields:

$$\Delta P = P_1 - P_2 = \frac{1}{2}\rho (V_2^2 - V_1^2) \quad (11)$$

Substituting the exact continuity geometric boundary condition ($V_2 = 7.2485 \cdot V_1$) allows the pressure equation to form an isolated function of the initial entry velocity V_1 :

$$\Delta P = \frac{1}{2}(1.225) [(7.2485 \cdot V_1)^2 - V_1^2] \quad (12)$$

$$\Delta P = 0.6125 [52.5408 \cdot V_1^2 - V_1^2] = 0.6125 [51.5408 \cdot V_1^2] \approx 31.5687 \cdot V_1^2 \quad (13)$$

Numerical Verification and Energy Balances

Evaluating this pressure transformation function utilizing the exact input velocity $V_1 = 0.5814$ m/s:

$$\Delta P = 31.5687 \cdot (0.5814 \text{ m/s})^2 \approx 10.671 \text{ Pa} \quad (14)$$

This reveals that the fan mechanism must consistently overcome a minimum static pressure bottleneck of 10.671 Pascals to drive mass transport effectively through this contraction profile.

Conversely, if an engineering system implements a physical differential manometer sensor to measure this localized baseline drop ($\Delta P = 10.671$ Pa), the exit velocity parameter can be cleanly extracted without relying on known initial volumetric flow data:

$$V_2 = \sqrt{\frac{2 \cdot \Delta P}{\rho \left(1 - \left(\frac{A_2}{A_1}\right)^2\right)}} = \sqrt{\frac{2 \cdot 10.671}{1.225 \left(1 - \left(\frac{1}{7.2485}\right)^2\right)}} \approx 4.2145 \text{ m/s} \quad (15)$$